

GNN-Based BERT for Understanding Context from Music

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Abstract

Music contains rhythmic, harmonic, and structural information in audio, as well as semantic information in text such as genre, mood, and instrumentation. This project examines whether these two modalities can be combined to understand music context, through four tasks built on DistilBERT, GraphSAGE, CNNs, fusion models, and contrastive learning, using MusicCaps, FMA-small, and MagnaTagATune. DistilBERT achieved a test Micro-F1 of 0.7532 and Macro-F1 of 0.6475 on multi-label text classification, while GraphSAGE reached 25.00% genre accuracy versus 38.75% for a mel-spectrogram CNN. In multimodal fusion, BERT alone outperformed all GNN-BERT combinations (Macro-F1 0.5169), and simple concatenation (0.4657) outperformed cross-attention fusion. A contrastive audio-text retrieval model achieved a caption-to-audio Recall@10 of 0.5185. Overall, textual information proved most informative for music labels, while graph-based audio representations captured structure but did not surpass stronger text or CNN baselines.

1. Introduction

Understanding music is not just about recognizing sounds. A music clip has many features, genre, mood, tempo, instrumentation, recording style, and more. Some of these features can be detected directly from the audio. Others are easier to describe using words. Because of this, a model that truly understands music context can benefit from both audio and text data.

Traditional audio models usually work on waveforms or spectrograms. Convolutional neural networks are good at finding patterns in these representations. Music also has relationships between different parts of a track: a section might

repeat later, two parts might share harmony, and neighboring segments might form a progression. A graph is a natural way to represent these connections. Graph Neural Networks can learn from this kind of structure. By treating each time segment as a node, a GNN shares information between connected segments. GraphSAGE (1) fits well here because it aggregates information from neighboring nodes to build a representation of the whole graph.

Text gives another way to look at music. Captions, tags, artist names, album details, and descriptions can include clues about meaning. Transformer-based language models such as BERT (2) are able to understand the meaning of text. For this project, DistilBERT (3) was selected as the text encoder, it is smaller and faster than BERT while retaining much of its language understanding ability.

The main goal of this work was to explore these two sources of information step by step. The project was split into four tasks. Task 1 used text with a BERT-based multi-label classifier. Task 2 looked at music structure graphs using GraphSAGE and compared them to a CNN baseline that worked on audio alone. Task 3 combined the GNN and BERT outputs and tested different ways to fuse them. Task 4 shifted from classification to contrastive learning, training audio and text encoders to work in the same embedding space.

This task-by-task approach was helpful. It let us study how each type of data contributed on its own before combining them, and made it easier to see where the models did well and where the current representations fell short.

2. Datasets and Exploratory Analysis

Three datasets were used across the experiments: MusicCaps, FMA-small, and MagnaTagATune.

2.1. MusicCaps

MusicCaps comprises 5,521 musical samples featuring natural-language descriptions. Every row includes a list of aspects describing musical features. This dataset was used in Task 1 and Task 4. A preliminary examination revealed that all fields used in this project were complete, with no duplicate YouTube IDs. The captions had an average of 49 words; the shortest caption consisted of 3 words, while

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the longest contained 136 words.

The raw aspect labels featured overlapping phrases, for instance, terms such as *voice*, *male vocalist*, and *male singing* all refer to almost the same characteristic. Comparable overlaps emerged for instrument and tempo tags. The values were standardized before choosing labels for Task 1. Thirty standardized labels were selected, including characteristics like *medium tempo*, *bass guitar*, *fast tempo*, *male vocal*, *emotional*, *energetic*, *slow tempo*, *electric guitar*, *rock*, and *pop*. After excluding instances lacking any of these labels, 4,888 MusicCaps entries were left for Task 1. The data was split randomly using a fixed seed of 42, producing 3,910 training, 489 validation, and 489 test samples.

2.2. FMA-small

The FMA dataset (4) was used for genre analysis. The FMA-small subset consists of 8,000 tracks distributed among eight genres: Hip-Hop, Pop, Folk, Experimental, Rock, International, Electronic, and Instrumental. Each genre contains 1,000 tracks, making FMA-small naturally balanced for genre classification. For Task 2, a reduced balanced subset was selected while maintaining the FMA partition: each genre provided 200 training tracks, 50 validation tracks, and 50 test tracks, for totals of 1,600 training, 400 validation, and 400 test tracks. This ensures a fair comparison between the Graph Neural Network and the Convolutional Neural Network, so that neither is overshadowed in the assessment.

2.3. MagnaTagATune

MagnaTagATune (5) contains audio clips with a large set of music tags. The metadata used in this project contained 188 tag columns. Task 3 focused on ten labels: genre-related (rock, pop, classical, techno, electronic, ambient) and context/mood-related (slow, fast, loud, quiet).

The metadata included 14,322 clips with at least one of these ten labels. Due to computing and storage restrictions, 1,200 examples were selected using a fixed seed of 42.

The textual representation for Task 3 was created from metadata associated with each MagnaTagATune clip. It contained the title, artist, album, and up to twelve positive tags that were not part of the ten prediction targets, preventing target-label leakage into the text input. For example: “*Title: Bisonpolska. Artist: Barry Phillips. Album: Trad. Musical characteristics: violins, strings, classic, violin.*”

The standard folder-based split from MagnaTagATune was used. Prior to graph generation, the subset contained 870 training, 76 validation, and 254 test instances. One training audio clip failed during preprocessing, so the final graph dataset comprised 869 training, 76 validation, and 254 test graphs.

3. Preprocessing and Graph Construction

3.1. Audio preprocessing

Audio was imported as mono and resampled to 22,050 Hz. For graph-based models, audio was divided into 5-second segments, and a 12-bin chroma representation was computed for each segment, capturing energy across the 12 pitch classes. The average chroma vector per segment ensured uniform-length features. For the CNN approach in Task 2, 10 seconds of audio was converted into a 128-bin log-mel-spectrogram, resized for uniform dimensions.

3.2. Segment graph

Every valid audio segment became a graph node with a 12-dimensional chroma feature vector. Two kinds of edges were used: temporal edges connecting consecutive segments bidirectionally, and similarity edges connecting segments whose chroma vectors were highly alike (cosine similarity threshold of 0.997). The resulting graph thus captured both temporal organization and links between musically similar segments.

3.3. Text preprocessing

DistilBERT was used for each text encoder, with inputs handled by the `distilbert-base` tokenizer. For Task 1 and Task 3, the maximum sequence length was 128 tokens; longer inputs were truncated, and shorter inputs were padded. Task 1 used natural captions from MusicCaps; Task 3 used the semantic metadata text generated from MagnaTagATune.

4. Task 1: BERT-based Music Tag Classification

4.1. Model

The pretrained `distilbert-base-uncased` model was used with an added 30-output classification layer, one per music attribute, treated as multi-label classification. For a text input x , DistilBERT produced a representation t , and the classifier computed a probability for each label k as:

$$\hat{y}_k = \sigma(w_k^T t + b_k) \quad (1)$$

The model was trained with binary cross-entropy loss using AdamW at a learning rate of 2×10^{-5} , batch size 16, for 5 epochs. At test time a threshold of 0.5 converted probabilities into binary predictions.

4.2. Results

The final Macro-F1 of 0.6475 shows the model was not simply learning to predict the most frequent labels. Task

Table 1. Validation Micro-F1 and Macro-F1 for Task 1 across five training epochs.

Epoch	Val. Micro-F1	Val. Macro-F1
1	0.2966	0.0927
2	0.5643	0.3509
3	0.6543	0.4780
4	0.7008	0.5507
5	0.7413	0.6350

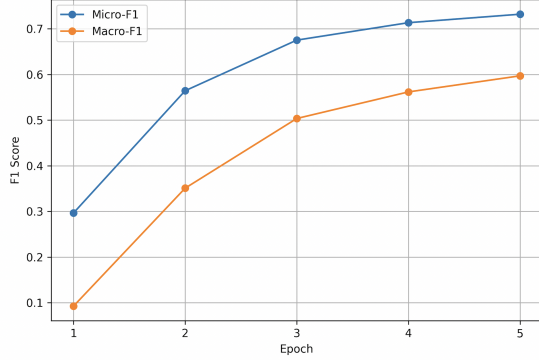


Figure 1. Validation Micro-F1 and Macro-F1 across five training epochs for Task 1.

1 provided the best classification result in the project, MusicCaps captions frequently describe tempo, instruments, vocals, recording quality, and mood directly, making text a highly informative source for these labels.

5. Task 2: Graph-based Genre Classification

5.1. GraphSAGE model

The GraphSAGE model used two message-passing layers (12→64, 64→64 dimensions) with ReLU activations, followed by global mean pooling and a linear classifier over eight genres. Trained with batch size 32, Adam at learning rate 0.001, for 20 epochs. Training loss decreased from 0.4790 to 0.3573; validation accuracy rose from 12.50% to 24.25%.

5.2. CNN baseline

A baseline CNN on log-mel spectrograms used three convolutional layers (channels 16, 32, 64) with max pooling, adaptive average pooling to 4×4 , a 128-unit fully connected layer, and an eight-way softmax. Trained for 10 epochs with Adam ($\alpha = 0.001$), cross-entropy loss, batch size 16.

5.3. Results

Both models beat the 12.50% chance baseline, but the CNN decisively outperformed GraphSAGE. This gap largely

Table 2. Task 1 test set results.

Metric	Score
Micro-F1	0.7532
Macro-F1	0.6475

Table 3. Task 2 test accuracy comparison.

Model	Test Accuracy
Random chance	0.1250
GraphSAGE	0.2500
CNN (mel-spectrogram)	0.3875

stems from feature bottlenecking: 12 chroma bins capture harmony but discard rhythm, timbre, and percussion, all essential for genre identification, whereas the CNN benefits from full mel-spectrogram detail. This does not mean graph-based approaches cannot work for audio — richer node features (MFCCs, learned embeddings, rhythmic descriptors) would give the graph model a fairer comparison.

6. Task 3: GNN-BERT Fusion for Music Context

6.1. Experimental setup

Four models were compared: BERT-only, GNN-only, Concat (GraphSAGE + DistilBERT vectors concatenated), and Cross-attention (graph and text projected to a shared dimension and combined via multi-head attention). All predicted the ten MagnaTagATune labels. The GNN encoder had two 64-dimensional GraphSAGE layers; concatenation joined the 64-dim graph embedding with the 768-dim text embedding into an 832-dim representation fed to a 256-unit hidden layer; cross-attention projected both to 256 dimensions with four attention heads. Each model trained for three epochs (GNN lr 0.001, BERT components 2×10^{-5}), with thresholds chosen on validation.

6.2. Ablation results

BERT-only achieved the best scores across all metrics; concatenation was the best model using both modalities. Adding an additional modality did not inherently improve predictions: the semantic input already contained strong genre hints (title, artist, album, non-target labels), while the graph side used only normalized chroma from 5-second intervals. Cross-attention underperformed concatenation, likely due to its added parameters and the relatively small training set (869 graphs).

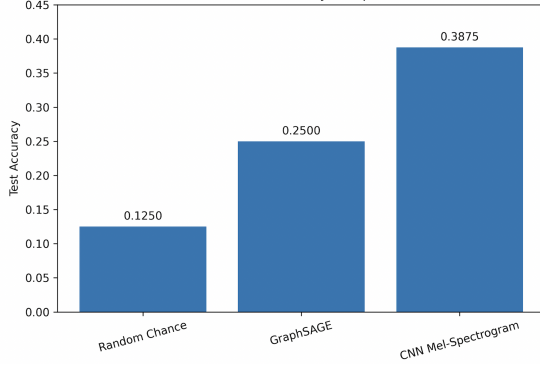


Figure 2. Task 2 test accuracy comparison between random chance, GraphSAGE, and the CNN baseline.

Table 4. Task 3 ablation results across fusion strategies.

Model	Thr.	Micro-F1	Macro-F1	AUC-PR
BERT-only	0.23	0.5964	0.5169	0.6096
GNN-only	0.05	0.2746	0.2670	0.2180
Concat	0.20	0.5430	0.4657	0.5595
Cross-attn.	0.17	0.3795	0.2572	0.3360

6.3. Embedding visualization

A 256-dimensional fused representation was extracted from the cross-attention model for test cases and projected to 2D via t-SNE (6), colored by genre/mood tag. Some same-tag instances cluster together, but clusters are not fully distinct, consistent with the metrics, music labels frequently overlap (e.g., a clip can be both electronic and ambient).

6.4. Case studies

Three test cases were examined: *Skitzo Prom Night* (metadata: “metal”; target: “rock”); *Deluxe* by Strojovna 07 (metadata: “electronica”; target: “techno”); and *Nightheat* by Shiva In Exile (target: “rock”). Attention weights over text tokens were relatively even rather than sharply focused, helping explain why cross-attention did not exceed concatenation — it failed to establish selective alignment between graph structure and specific semantic tokens.

7. Task 4: Cross-modal Audio-Text Retrieval

7.1. MusicCaps subset

A reproducible random held-out split (unlike Tasks 2–3, determined by audio availability) yielded 212 training, 27 validation, and 27 test pairs.

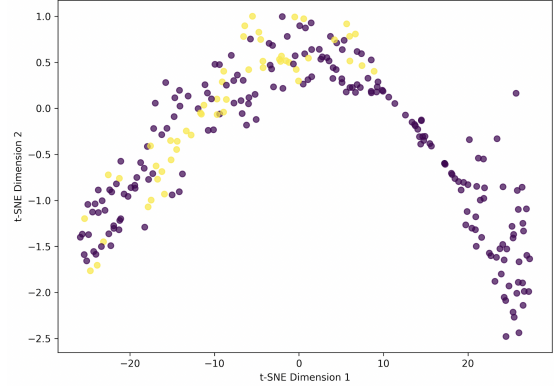


Figure 3. t-SNE visualization of the fused Task 3 embedding space.

Table 5. Retrieval results for Task 4, in both directions.

Direction	R@1	R@5	R@10
Caption to Audio	0.0000	0.2963	0.5185
Audio to Caption	0.0370	0.1481	0.4444

7.2. Dual-encoder model

The audio encoder used two 128-hidden-unit GraphSAGE layers with global mean pooling and a linear projection to a 128-dimensional embedding; the text encoder used DistilBERT projected to the same space. Both embeddings were L2-normalized. The contrastive loss combined audio-to-text and text-to-audio cross-entropy terms:

$$\mathcal{L} = \frac{\mathcal{L}_{a \rightarrow t} + \mathcal{L}_{t \rightarrow a}}{2} \quad (2)$$

with temperature 0.07, trained for five epochs using AdamW at 2×10^{-5} . Training loss decreased steadily from 2.7020 to 2.2486, showing the model learned to align paired audio and text.

7.3. Retrieval results

Caption-to-audio retrieval was effective at R@5 and R@10, over half of queries located their match among the top ten results. R@1 remained low, likely due to the small test set (27 pairs) and limited training pairs (212). Chroma-only audio features also cannot capture vocal, timbral, or production cues a caption may describe. Nonetheless, R@10 results indicate a meaningful learned relationship between modalities.

7.4. Qualitative retrieval example

To inspect the retrieval behavior beyond Recall@K, one caption query was examined together with its top three retrieved audio clips. The query described an amateur recording con-

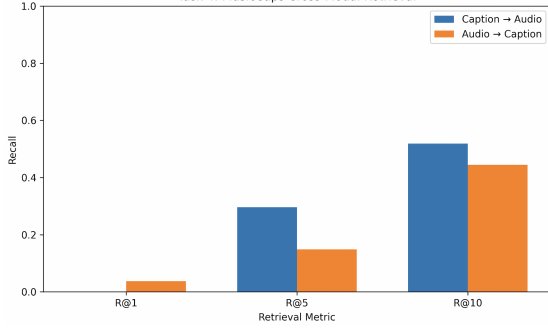


Figure 4. Task 4 Recall@1, Recall@5, and Recall@10 for caption-to-audio and audio-to-caption retrieval.

Table 6. Example caption-to-audio retrieval from Task 4.

Rank	Retrieved YouTube ID	Similarity
1	mtapmFDmImA	0.1188
2	lh6ACdwNlyk	0.0643
3	PE1ges9nn6A	0.0492

taining gargling sounds, female voices, background pop music, and fast popping noises.

None of the top three clips was the exact paired audio for this query. This example is consistent with the overall retrieval results, where Recall@1 was low while Recall@5 and Recall@10 were noticeably better. It also shows that the learned embedding space could rank candidate clips, but fine-grained audio-text alignment remained difficult with the small training set and chroma-only audio representation.

7.5. Zero-shot tag experiment

Tag-name text embeddings were compared against the learned multi-modal representation. This cannot be directly compared to Task 3 F1 values since Task 4 was never trained with those label tags; no numerical Macro-F1/AUC-PR is reported. The contrast still reveals two complementary settings: Task 3 learns from supervised labels, while Task 4 aligns audio with text and can apply text concepts without direct label supervision.

8. Discussion

8.1. Comparison across tasks

The four tasks show different strengths of text, audio, and multimodal learning. Task 1 gave the strongest classification result because the MusicCaps captions directly describe properties such as mood, tempo, instruments, and vocals. Task 2 showed that graph-based audio features can capture useful structure, but the GraphSAGE model was weaker than the mel-spectrogram CNN. This suggests that the 12-dimensional chroma representation removed important in-

formation such as timbre and rhythm.

Task 3 showed that combining modalities does not automatically improve performance. BERT-only achieved the best result, while simple concatenation performed better than cross-attention. This indicates that a more complex fusion method needs enough data and strong features from both modalities to be useful.

Task 4 moved beyond classification and tested whether audio and text could share the same embedding space. The model achieved better Recall@5 and Recall@10 than Recall@1, showing that it learned some useful alignment but still struggled to identify the exact matching clip at the top rank. Overall, the experiments suggest that data quality and feature quality were more important than model complexity in this project.

The major limiting factor across experiments is data volume: Task 3 used 1,200 MagnaTagATune instances (one failed during graph processing); Task 4 had only 266 complete audio-caption pairs, low figures for models built on pretrained transformers and GNNs.

Graph node representation is also limiting: chroma captures pitch-class energy but not timbre, lyrics, singer identity, or percussion. Future work could integrate MFCCs, tempo attributes, or pretrained audio embeddings. Graph construction (fixed 0.997 cosine threshold) could also be improved via learned or adaptive edge weighting.

Task 3 could benefit from more MagnaTagATune data, particularly for cross-attention, whose parameters are difficult to train at this scale. Task 4 would benefit from more audio-caption pairs and hard negative examples. Evaluation of Task 4 also requires human assessment (at least five raters scoring retrieved-clip relevance 1–5); this has not yet been collected.

Overall, the results support that multimodal models are not automatically better simply because they contain more components, the quality of information entering each component matters more than model complexity.

Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning, specifically multimodal music understanding. The methods and datasets used (MusicCaps, FMA-small, MagnaTagATune) are publicly available research datasets, and no personally identifying information about individuals is processed. There are many potential societal consequences of this work, none which we feel must be specifically highlighted here.

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